A Fast Progressive Ship Detection Method for Very Large Full-Scene SAR Images

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***Abstract*— Synthetic aperture radar (SAR) has emerged as a vital tool for ship monitoring due to its all-weather, all-day, high-resolution imaging capabilities. In practical operations, the wide coverage and sparse ship distribution in very large full-scene SAR images pose challenges in terms of low efficiency and high false alarm rates. Traditional methods perform poorly in complex scenarios, while deep learning (DL) methods have high computation costs. This study proposes a fast progressive detection algorithm for ship targets in large SAR images, combining the advantages of traditional Non-DL methods and DL approaches. First, at a global scale, image preprocessing operations based on traditional methods are designed to quickly extract candidate regions. Then, at the regional scale, an oriented ship detector (OSD) is designed for refined ship detection within candidate regions. Finally, at an individual-target scale, a false alarm discrimination network is constructed to further remove false alarms. Experimental results on GF-3 full-scene SAR images demonstrate that the proposed method can achieve minute-level detection efficiency in images of billion-pixel-level size while achieving high detection accuracy.**

***Index Terms*— Candidate region retrieval, progressive detection, ship detection, synthetic aperture radar (SAR), very large full-scene SAR images.**

I. Introduction

With the advancement of earth observation and satellite imaging technologies, remote sensing images exhibit characteristics of wide swath and high resolution, both expanding the coverage area and enriching the target details. Automatic target detection technology liberates users from manually exploring massive image data. Compared to optical sensors, SAR has the unique advantages of all-weather and all-day monitoring. In recent years, a plethora of research outcomes on target detection based on SAR images has emerged, focusing on targets like aircraft, ships, and vehicles. Among them, ship detection is useful in fishery management, port planning, and marine monitoring. This article primarily focuses on ship target detection in nearshore and offshore areas on a single channel, very large full-swath SAR images, which are of a billion-pixel-level size.

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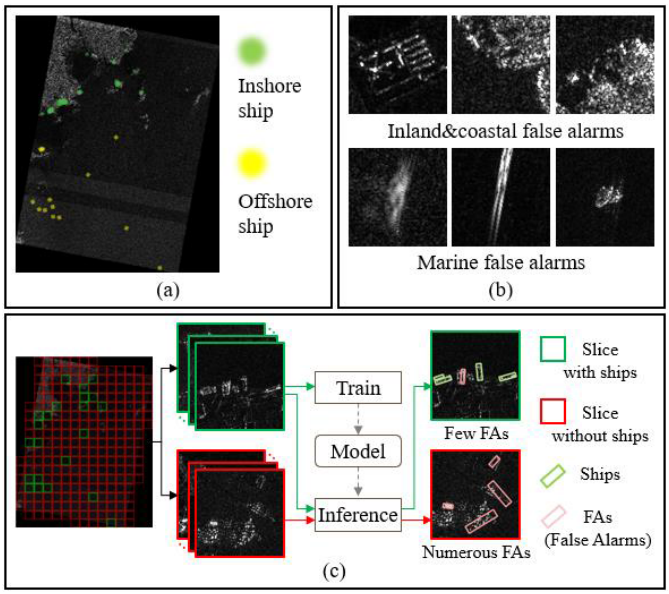

Fig. 1. Challenges faced in ship target detection within large scene SAR images. (a) Sparse distribution of ships. (b) Common false alarms in SAR ship detection. (c) Causes of false alarms associated with DL-based methods.

In practical applications, SAR images are often presented to users in entire scenes, posing two challenges for ship target detection: low efficiency in processing very large coverage and high false alarm rate in detecting sparse targets in complex backgrounds. As illustrated in Fig. 1, these challenges manifest in the following three aspects.

A. Large Scene With Sparse Ships

Full-scene SAR images cover a very large area with sparse and uneven ship distribution. Fig. 1(a) shows the distribution of ships in an SAR image, where green dots indicate nearshore ships concentrated in areas like ports, docks, and river estuaries, and yellow dots represent offshore ships with much sparser distribution. The low proportion of ship targets within the image suggests that there are only very few valid regions worth detecting, and thus, rapid localization of the candidate regions is one of the key issues in enhancing detection efficiency.

B. False Alarms in Non-DL Methods

Product-level single-channel SAR images retain only amplitude information, which, compared to multichannel optical images, provides less information; moreover, SAR images

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are affected by geometric distortion and speckle noise, along with the interference from complex environments, such as coastlines, islands, ports, and sea clutter, causing traditional nondeep learning (non-DL) methods often produce a great number of false alarms [1]. Fig. 1(b) exhibits some common examples of false alarms, including man-made facilities, coastal areas, sea clutter, and small islands.

C. False Alarms in DL-Based Methods

Contrary to traditional methods, DL-based methods exhibit superior false alarm suppression; however, for ship detection tasks of large-scene SAR images, the issue of high false alarm rates still persists, primarily appearing in the untrained regions. Fig. 1(c) explains the reason for false alarms in DL-based approaches. Because of GPU computational limitations, DL detectors divide large-size images into small scene slices for training. To maintain a balance between positive and negative samples, those slices containing targets are mainly used during the training process, as depicted by the green slices in Fig. 1(c); however, many more slices are without any target, illustrated as red slices in Fig. 1(c), but contain numerous negative samples (false alarms) yet remaining untrained. Subsequently, during the detection stage, the model is required to perform inference across the entire image, wherein these untrained regions contribute to a significant portion of false alarms.

Current SAR ship detection algorithms can be mainly divided into traditional non-DL and DL-based methods. Traditional methods, represented by constant false alarm rate (CFAR) and its variants [2], use statistical modeling of sea clutter and false alarm rate thresholds for ship detection. These methods often yield numerous false alarms and have limited generalization capability to new scenarios. Enhancements like multiscale detection [3], [4] improve accuracy but require complex parameter tuning. On the other hand, DL-based methods, widely applied in optical remote sensing target detection, are also adopted in SAR image target detection [5]. These methods, driven by extensive annotated data, offer higher accuracy and better generalization. Hardware constraints, however, limit the image size of network input, necessitating the use of sliding windows for detection, which severely slows down the detection process.

To address the challenges and limitations of existing methods, this article proposes a fast and progressive ship detection algorithm in very large full-scene SAR images to achieve a good balance between speed and accuracy. It contains two major modules: one for the retrieval of candidate ship regions and another for high-precision ship detection, integrating the advantages of both traditional and DL-based methods. To solve the problem of low efficiency in large-scene detection, a global-scale preprocessing strategy inspired by traditional approaches is employed for fast extraction of candidate regions. Then, an evaluation metric is devised for the evaluation of the candidate region retrieval module (CRM), ensuring effective background rejection and reliable foreground extraction. In the subsequent stage, DL is used to formulate an oriented ship detector (OSD) and a false alarm discrimination module (FDM). OSD is operated in a sliding window fashion to enhance detection accuracy, and FDM focuses on minimizing false alarms on target scales. The main contributions of this article are summarized as follows.

1) An end-to-end fast detection method based on a progressive detection approach is proposed for ship targets in a very large full-scene SAR imagery. The method enhances the efficiency of detection on full-scene images and increases detection precision.

2) A ship CRM tailored for large-scene SAR imagery is designed to quickly locate regions of interest, reducing the scope for fine detection, and its effectiveness is evaluated with a specially devised performance metric.

3) A high-precision detection module is designed for SAR ship targets and consists of OSD and FDM that operate at region and target scales, respectively, for refined detection and false alarm suppression, ensuring accurate detection within candidate areas.

4) The proposed method is experimentally validated on GF-3 full-scene images, and the results indicate the method's capability to achieve minute-level detection efficiency on billion-pixel-scale images while maintaining high accuracy.

The remainder of the article is organized as follows. Section II reviews related works, Section III presents the theoretical and technical details of the proposed method, Section IV discusses experimental results, and Section V summarizes the article.

II. Related Works

A. Traditional Methods

SAR automatic target recognition (ATR) is an important technology for the automatic interpretation of SAR imagery. Since the 1980s, the three-staged methodology for SAR ATR developed by Lincoln Laboratory [6], including detection, discrimination, and classification, has been widely used. Algorithms devised for ship target detection in the SAR domain are, to a significant extent, derivatives of this foundational structure, particularly emphasizing the detection and discrimination phases.

The initial phase of SAR ATR involves CFAR detection, which is based on clutter statistics and threshold determination. The two-parameter CFAR detector uses a Gaussian clutter distribution to adaptively determine thresholds for target detection [2]. Subsequently, a variety of improved CFAR algorithms has emerged, dealing with clutter modeling [7], detector design [8], and algorithmic efficiency [9]. In terms of SAR ship detection, Leng et al. [10] proposed a bilateral CFAR leveraging kernel density estimation for enriched spatial analytics. Dai et al. [11] substitute the static sliding window with candidate regions as protective buffers, mitigating multiscale ship detection lapses inherent in traditional CFAR. Cui et al. [12] employ neural networks for clutter modeling in CFAR, forecasting segmentation thresholds and thereby enhancing detector performance under complex conditions.

In the discrimination phase of SAR ATR, it is important to suppress high false alarm rates. Conventional discriminators harness the textural properties of targets to distinguish

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them from clutter such as support vector machines [13]. Liu et al. [14] employ ship size and contextual information combined with binary linear programming. Wang et al. [15] model complex maritime clutter using aggregated Fisher vector densities; moreover, neural networks have also been deployed for distinguishing ships from false alarms [4].

B. DL-Based Methods

1) Oriented Detectors: As targets in remote sensing images often have arbitrary orientations, the use of oriented bounding boxes (OBBs) is popular due to their ability to more accurately represent targets, especially those with high aspect ratios like ships. The RoI Transformer (RoI-Trans) [16] initiated the trend of two-stage oriented detectors. Then, oriented R-CNN [17] and ReDet [18] further refined this concept by simplifying network complexity and introducing rotation-invariant features, respectively. In terms of single-stage detectors, R3Det [19] is built upon RetinaNet [20] to enhance oriented regression and classification through cascading detectors and feature refinement. SCRDet++ [21] augmented this approach by integrating object-level attention modules. Others have focused on optimizing OBB representation to sharpen localization accuracy [22], [23].

2) Tailored Detectors for SAR Ship Targets: To address practical application needs, a variety of DL-based detectors tailored for SAR ship targets have emerged. Li et al. [24] introduced Faster R-CNN to SAR ship detection. Cui et al. [25] proposed a shuffle-group enhanced attention module to extract stronger semantic features of ships and suppress noise. Sun et al. [26] improved near-shore SAR ship detection by identifying strong scattering points on ships and using attention modules to suppress land interference. Ke et al. [27] developed an improved, faster R-CNN algorithm that employs deformable convolutional kernels for SAR ship detection. He et al. [28] presented an anchor-free ship detection method that combines adaptive feature encoding with Gaussian-guided sampling.

Considering the exceptional detection capabilities of DL-based methods, this article introduces a high-precision ship detection module based on DL, enhanced by feature fusion strategies to further improve detection accuracy.

C. Ship Detection in Large-Scene Images

Several methods have been developed for ship detection in large-scene SAR images. Cui et al. [25] enhanced large-scene detection efficiency by incorporating a coarse sea-land segmentation module, which also reduced false alarms from land regions; however, this module lacked an in-depth analysis. In [26], [29], and [30], the oriented detectors were improved by integrating strong scattering points from SAR ships, enhancing detection capabilities in complex scenes. Yet, for large-scene detection, they still relied on full-image slicing, with limited exploration of detection efficiency. Conversely, Zhang et al. [31] used multiscale detection to increase recall and designed two postprocessing strategies to reduce the false alarm rate in large-scene detection. Shao et al. [32] used CFAR to narrow down the detection scope and enhance the efficiency of subsequent DL-based detectors. Additionally, Wang et al. [33] proposed a network lightweight method to speed up model loading and execution for ship detectors.

As described in Fig. 1, large-scene SAR ship detection in practical applications primarily faces challenges of low detection efficiency and high false alarm rates. Among the mentioned methods, some focus mainly on optimizing DL-based detectors, such as [26], [29], [30], and [33], without thoroughly investigating preprocessing methods crucial for the efficiency of large-scene detection tasks. While [25] and [32] introduced preprocessing for the removal of land and sea clutter, they did not consider both aspects simultaneously. Zhang et al. [31] offered effective postprocessing strategies to deal with high false alarms but did not incorporate preprocessing to enhance detection efficiency. Compared to these existing approaches, this article provides a more systematic solution to enhance practicality in real-world applications. The method combines preprocessing, detector optimization, and postprocessing. The preprocessing phase considers both land and sea clutter removal to improve overall detection efficiency, while postprocessing further suppresses false alarms generated during detection.

III. Methodology

A. Progressive Efficient Detection Framework

The proposed method's overall framework, as depicted in Fig. 2, includes three steps corresponding to the introduced CRM, OSD, and FDM. These three submodules operate progressively across three scales, from coarse to fine, successively narrowing the processing scope to enhance detection performance.

The first step employs CRM for rapid candidate region retrieval at a global scale, incorporating sea-land segmentation and CFAR processing. It begins with sea-land segmentation to obtain a land area mask and exclude most terrestrial regions. Subsequently, CFAR processing is performed on the remaining sea surface to create a mask of the maritime area with ship targets to remove the majority of sea clutter. This preprocessing step aims to quickly locate candidate ship regions without DL-based detection and has minimal reliance on computational resources. It is, therefore, conducted at a global scale, and the processing speed is enhanced through image downsampling. This stage outputs a mask of the candidate regions.

The second step involves refined detection using OSD at the regional scene slice scale. Because of the limitations of GPU resources on which DL-based methods depend, the size of the image input of the network is restricted; thus, the sliding window strategy is employed for ship detection. The sliding window is based on the original image and is matched with the candidate region mask to determine whether the current window should be sent to the detector. The results from each window are mapped to full scene images, and nonmaximum suppression (NMS) is applied to the merged results to produce preliminary detection results.

The third step suppresses false alarms at the individual-target scale. For the preliminary detection results, chips of the target area above the confidence threshold are fed into a

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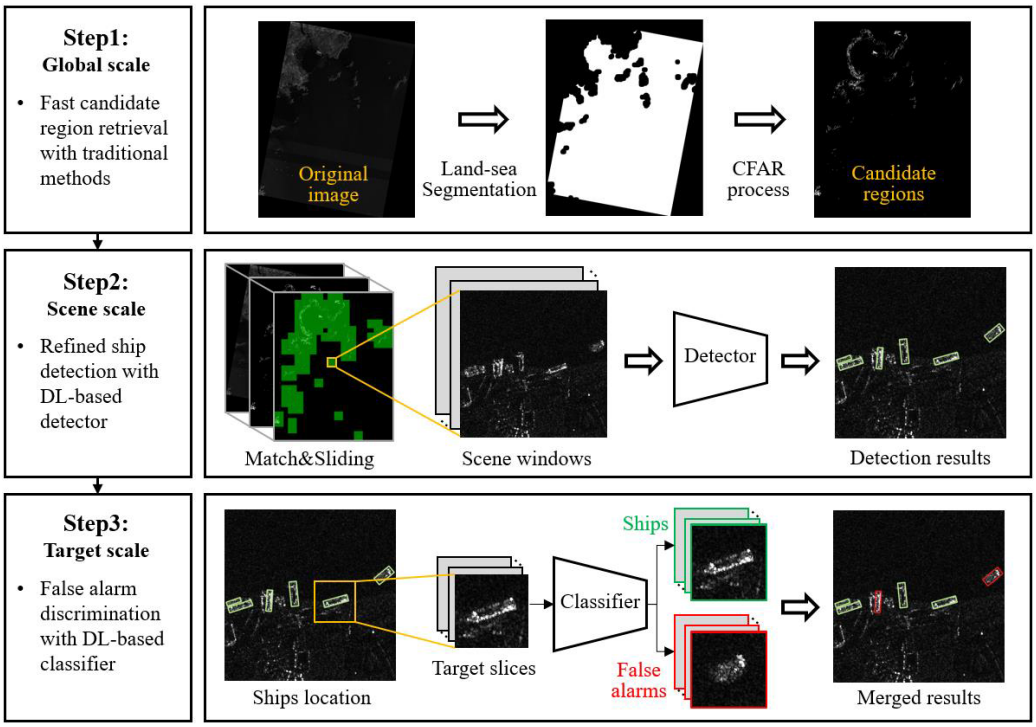

Fig. 2. Overall framework of the proposed method.

DL-based classifier for ship and false alarm discrimination. The classification results are restored to the scene image, and nonship categories are filtered out.

Additionally, the proposed framework leverages block-reading and sliding window strategy to enhance efficiency for large-scene images, as depicted in Fig. 3. The process contains multiple scales, from a broad global view to finer scene and target slices. Block reading is applied for localized image access at the scene and target levels, as shown in steps 2 and 3 in Fig. 3, facilitated by the use of GDAL [34] raster reading tools. After image preprocessing, the candidate region mask is saved as a binary image on disk, which guides the detection networks to focus only on candidate regions. Block reading improves efficiency by selectively reading smaller detection regions from external storage into memory, followed by quantization stretching before the detection process. The false alarm discrimination also retrieves target slices similarly, isolating and stretching individual slices for classification. Additionally, block processing supports final result slicing and concurrent saving using a multiprocess architecture.

The proposed framework employs sliding windows for block reading (W_{block}) and OSD (W_{det}). Each block slice is read as W_{block} into memory, stretched, and then detected through the W_{det}. To use the sliding windows most efficiently, W_{block} and W_{det} maintain the following relationship:

$$W_{\text{block}} = W_{\text{det}} + \left(\sqrt{BS} - 1\right)(W_{det} - OL). \tag{1}$$

In this equation, BS represents the batch size of the OSD and OL represents overlap.

The following three subsections will introduce the CRM, OSD, and FDM, respectively.

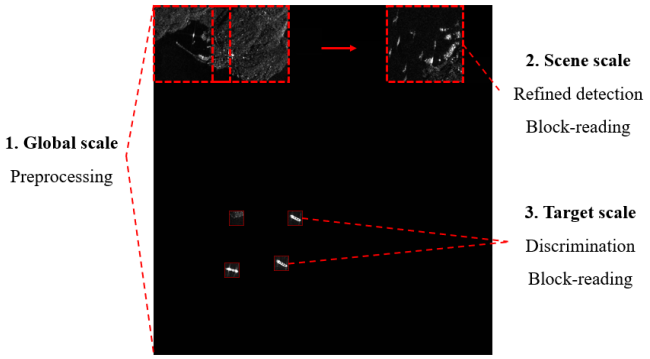

Fig. 3. Multiscale block reading and sliding windows.

B. Candidate Region Retrieval Module (CRM)

1) Land Area Exclusion Based on Sea-Land Segmentation: In the proposed framework, sea-land segmentation is employed as a preprocessing module. Its design is primarily determined by two factors: the execution efficiency and the proportion of nearshore ships retained after segmentation. The objective is to eliminate terrestrial regions as much as possible without imposing strict constraints on the precision of segmentation.

a) Execution efficiency: Conventional sea-land segmentation methods have two ways: geographic information-based matching and image processing. Image processing approaches suffer from slower execution times, particularly when applied to full-scene SAR imagery. Given that the images under analysis in this study are geocoded, the proposed method opts for the swifter geographic information-based template matching. To enhance the speed of template reading and matching, a global sea-land mask comprised of GeoTIFF

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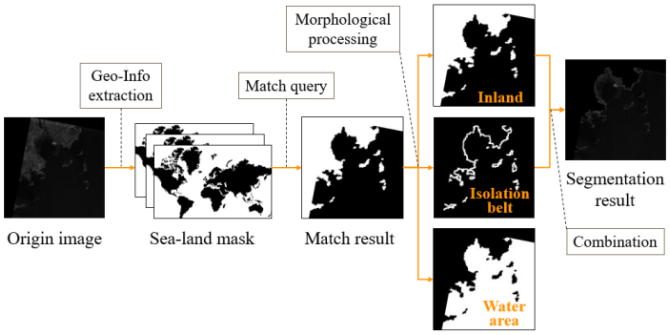

Fig. 4. Sea-land segmentation based on mask matching.

raster tiles is employed as a repository with a resolution of 250 m [35].

b) Ship preservation: Geocoded remote sensing images often exhibit certain offsets, the phenomenon particularly pronounced in SAR imagery. This can lead to inaccurate matching, erroneously eliminating nearshore ships. To mitigate this issue, morphological erosion is incorporated to contract the coastline, creating a sea-land isolation belt to maximally preserve nearshore ship areas. The hyperparameters for the operation are evaluated and adjusted based on 20 scenes of GF-3 data. This adjustment is made considering both the proportion of ship preservation and the proportion of land elimination, with specific metrics detailed in the latter part of the "preprocessing evaluation metrics design."

As depicted in Fig. 4, the geographic information-based segmentation begins with the extraction of geographical metadata from the original image, followed by the retrieval of the corresponding mask from the database. This mask is aligned and segmented against the original image to produce an initial result. The segmentation is refined through morphological processing, with an erosion followed by a dilation operation on the sea-land boundary, resulting in separate zones: inland, isolation belts, and oceanic regions. The inland and isolation zones are merged to form the definitive sea-land segmentation output, and the oceanic region is reserved for the sea clutter filtering stage.

2) Sea Clutter Filtering Based on CFAR: This section introduces a CFAR-based sea clutter filtering approach, which includes steps of coastal line dilation, clutter modeling, and coastal line postprocessing.

a) Coastal line dilation: Because of the use of template matching in sea-land segmentation, the extracted maritime area still includes some coastal land regions influenced by geographical encoding deviations. These areas have the highest pixel intensity, introducing uncertainties in modeling sea clutter. To better model the sea clutter region, a morphological erosion is, therefore, applied to the coastline delineated by the sea-land segmentation. This step is performed after the removal of the sea-land isolation belt to fully eliminate any terrestrial interference with the statistical modeling.

b) Clutter modeling: When fitting the sea clutter pixel regions, it is necessary to exclude artifacts, such as ships, which typically exhibit higher energy levels in SAR imagery. Referring to [3] and [4], a threshold of the pixel mean is employed to roughly eliminate ship-related pixels. Then, The study uses the generalized gamma distribution (GGD) for its proven efficacy in modeling SAR sea clutter [12]. The probability density function (pdf) of the GGD is defined as follows:

$$f(x;\alpha,\beta,\mu)=\frac{\beta^{\alpha}x^{\alpha\mu-1}e^{-\beta x^{\mu}}}{\Gamma(\alpha)},\ x\geq 0. \tag{2}$$

In the formula, f denotes the pdf, Γ represents the gamma function, and α, β, μ are the shape, scale, and energy parameters, respectively.

For practical applications, the segmentation threshold T of CFAR is derived by calculating the probability of false alarm (PFA) from the GGD-based pdf and cumulative density function (cdf), and PFA is defined as:

$$C(x;\alpha,\beta,\mu)=\int_0^x f(t;\alpha,\beta,\mu)dt,\ x>0 \tag{3}$$

$$P_{FA}=1-C(T;\alpha,\beta,\mu). \tag{4}$$

Here, C is the cdf, and P_{FA} quantifies the likelihood of sea clutter being misclassified as ships. Pixels with values above T are classified as ship targets. Adjusting the P_{FA} enables the determination of the most effective T, and this study adopts an iterative optimization approach [3], which is given in Algorithm 1. The iterative cycle concludes with the computation of the ideal T, deduced from the current P_{FA}.

Algorithm 1 Optimize CFAR Threshold

Input:
Image pixels without land, *Sea*
Initial probability of false alarm, P_{FA}
Adjustment parameter for P_{FA}, *delta*
Estimated ship count by image mean value, N_{ship}
Output:
Ship binary mask, $\boldsymbol{M}_{ship}$
CFAR threshold, T
Begin
$[p_1,\ p_2]$ = GammaFit(*Sea*)
T = ComputeGammaInv(1 - P_{FA}, p_1, p_2)
$\boldsymbol{M}_{ship}$ = SegmentShips(*Sea*, T)
n = CountObjects($\boldsymbol{M}_{ship}$)
$iter = 0$
While ($n >= N_{ship}$ OR $n <= 0$) AND $iter < maxIter$
If $n >= N_{ship}$
$P_{FA} = P_{FA} - delta$
Else
$P_{FA} = P_{FA} + delta$
End If
T = ComputeGammaInv(1 - P_{FA}, p_1, p_2)
$\boldsymbol{M}_{ship}$ = SegmentShips(*Sea*, T)
n = CountObjects($\boldsymbol{M}_{ship}$)
$iter = iter + 1$
End While
Return $\boldsymbol{M}_{ship}$, T
End

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c) Coastal line postprocessing: The proposed sea-land approach is not entirely precise, especially in the sea-land isolation belt, resulting in the retention of some terrestrial regions near the coastline. Following the establishment of the CFAR threshold, a threshold-based postprocessing is applied to the sea-land isolation belt. This step further eliminates pixels adjacent to coastal terrestrial regions but also preserves ship targets situated near the coastline within the sea-land demarcation zone. Additionally, this process directly uses the CFAR threshold, eliminating the need for additional sea clutter modeling and thereby having a negligible impact on the overall processing speed. Subsequently, a binary mask encompassing the entire area of candidate regions is produced.

3) Preprocessing Evaluation Metrics Design: It is imperative to ensure a high coverage rate of ship targets when reducing detection zones to accelerate the detection process by CRM, thereby minimizing the risk of false negatives. To evaluate the efficacy of the CRM, two evaluation metrics are developed: true hit rate and detection coverage.

a) True hit rate: The true hit rate, denoted by r_{hit}, is a metric for evaluating the proportion of ship targets covered in the candidate region after preprocessing. It is defined as follows:

$$r_{\text{hit}} = n_{\text{hit}}/n_{\text{total}}. \tag{5}$$

Here, n_{hit} is the number of ship targets hit in the candidate region, and n_{total} is the total number of true ship targets. A higher r_{hit} indicates fewer misses in the candidate region retrieval operation.

Whether an individual ship target is successfully hit by the candidate region is determined by the presence of candidate area pixels within its HBB. The candidate regions are depicted as binary masks in CRM, where 1 represents potential ship pixels and 0 indicates background. The illustration in Fig. 5(a) shows the target with its OBB annotation, and Fig. 5(b) reveals the HBB closely enveloping the OBB. In Fig. 5(b), gray pixels are where targets are filtered out after preprocessing, and red pixels indicate the retained candidate areas. A ship is considered detected if the candidate area contains any pixels within its HBB. Consequently, the computation of n_{hit} is as follows:

$$n_{\text{hit}} = \sum_{i=1}^{n_{\text{total}}} max(\boldsymbol{H}_i). \tag{6}$$

Here, $\boldsymbol{H}_i$ represents the pixel matrix of the ith target's HBB within the mask. The value of 1 in $\max(\boldsymbol{H}_i)$ indicates the presence of potential ship pixels.

The inclusion of ships within the subsequent networks is, furthermore, decided by the network's operational strategy. Using a sliding window approach, the network scans the entire original image, considering areas with pixels valued at 1 within the mask as potential ship presence. The presence of mask pixels within a ship's HBB ensures its coverage by at least one window, enabling its processing by the detection network. Edge cases, as illustrated in Fig. 5(c), show minimal preserved target pixels (in red), like just the stern, covered by a window edge. The design of overlapping areas in adjacent windows ensures complete target detection in such scenarios, validating the n_{hit} calculation's accuracy.

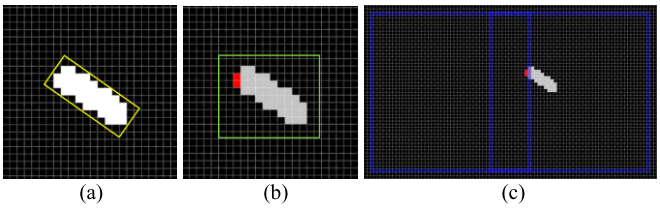

Fig. 5. Method to determine ship target hit by CRM. (a) OBB ground truth. (b) Post-CRM ship with its HBB ground truth. (c) Edge case of target preserved pixel and sliding window strategy.

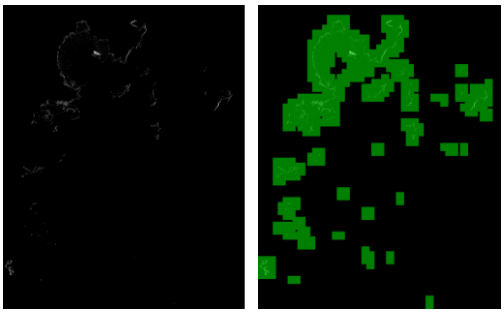

Fig. 6. Method to determine ship target hit by CRM. (a) OBB ground truth. (b) Post-CRM ship with its HBB ground truth.

b) Detection coverage: The detection coverage metric, represented as r_{cov}, evaluates the percentage of the entire scene image engaged by the detection network. A lower value of r_{cov} signifies enhanced speed in the subsequent detection phase. Analogous to r_{hit}, r_{cov} is derived from the processed candidate region mask of ship targets and hinges on the sliding window strategy employed by the detection network.

The illustration in Fig. 6, shows that the metric calculation involves emulating the sliding window procedure, considering the predefined window dimensions and overlap areas. Windows, including ship pixels in their corresponding mask regions, are classified as effective, denoted by green areas in the figure. The r_{cov} is quantified by the ratio of the original image's area occupied by the union of all effective sliding windows, which is defined as follows:

$$r_{\text{cov}} = \left(\bigcup_{i=1}^{n_w} a_i\right) \Big/ a_{\text{total}}. \tag{7}$$

Here, n_w represents the total number of effective sliding windows, a_i is the area of the current sliding window, and a_{total} is the area of the entire scene image.

C. Oriented Ship Detector (OSD)

1) Oriented Representation Methodology: In recent target detection, HBB and OBB are two predominant localization approaches. For elongated targets like ships, oriented detection algorithms have distinct advantages, including reducing the overlap of detection boxes for adjacent targets, efficient extraction of features in target areas, and providing richer semantic information.

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The definition of OBB significantly impacts the performance of oriented detectors. Long-side representation (LSR) is commonly employed for precise bounding box regression, but it suffers from issues of periodicity of angular (PoA). The midpoint offset representation (Mid-OR) [17], on the other hand, overcomes this challenge. It can, however, result in less precise bounding box regression, making it more suitable for initially extracting rotated regions of interest (RRoI).

The proposed method integrates a two-stage network, employing both Mid-OR and LSR for OBB. Specifically, Mid-OR is applied in the oriented region proposal network (RPN) [17] of the first stage to address PoA challenges. The box regression in this stage is formulated as follows:

$$\begin{aligned}
&\varphi_x = (x - x_a)/w_a, \quad \varphi_y = (y - y_a)/h_a \\
&\varphi_w = log(ww_a), \quad \varphi_h = log(h/h_a) \\
&\varphi_\alpha = \Delta\alpha/w, \quad \varphi_\beta = \Delta\beta/h \\
&\varphi'_x = (x' - x_a)/w_a, \quad \varphi'_y = (y' - y_a)/h_a \\
&\varphi'_w = log(w'/w_a), \quad \varphi'_h = log(h'/h_a) \\
&\varphi'_\alpha = \Delta\alpha'/w', \quad \varphi'_\beta = \Delta\beta'/h'.
\end{aligned} \tag{8}$$

In this equation, x and y represent the central coordinates of the box, while w and h denote its width and height, respectively. The terms x_a, x', and x correspond to the anchor box, ground-truth box, and predicted box, respectively, and similar notation applies to w, h, y, $\Delta\alpha$, and $\Delta\beta$. Subsequently, the regression loss is calculated using the smooth L1 loss, defined as:

$$L_{\text{reg1}} = \text{smooth}_{L1}(\varphi_i, \varphi'_i). \tag{9}$$

Here, φ_i and φ'_i denote the offsets of the predicted box and the ground truth box relative to the anchor box, respectively.

In the second stage, the approach retains the use of LSR to achieve greater precision in localizing bounding boxes. The box regression for this stage is articulated as follows:

$$\begin{aligned}
&t_x = (x - x_a)/w_a, \quad t_y = (y - y_a)/h_a \\
&t_w = log(w/w_a), \quad t_h = log(h/h_a), \quad t_\theta = \theta - \theta_a \\
&t'_x = (x' - x_a)/w_a, \quad t'_y = (y' - y_a)/h_a \\
&t'_w = log(w'/w_a), \quad t'_h = log(h'/h_a), \quad t'_\theta = \theta' - \theta_a.
\end{aligned} \tag{10}$$

In this formulation, θ_a, θ, and θ' represent the angles of the anchor box, ground-truth box, and predicted box, respectively. The other parameters are consistent with those in the first stage. The regression loss for this stage is also computed using the smooth L1 loss, which is defined as:

$$L_{\text{reg2}} = \text{smooth}_{L1}(t_i, t'_i). \tag{11}$$

Here, t_i and t'_i signify the offsets of the predicted box and the ground truth box in relation to the anchor box, respectively.

The second phase, inspired by RoI-Trans [16], projects the box's reference coordinate onto the RRoI, as shown in Fig. 7(a). Since the first stage has already approximated the target's angle, the projection operation ensures that most boxes requiring fine-tuning by the head network are not near critical values, as illustrated in Fig. 7(b), effectively circumventing PoA issues.

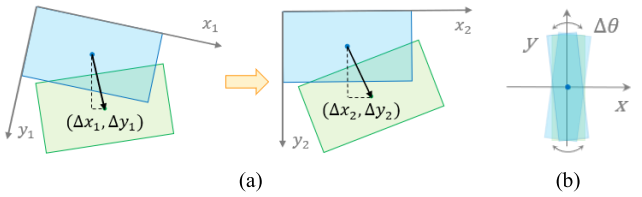

Fig. 7. Box regression process of head network, blue for boxes to be regressed and green for ground truth. (a) RRoI coordinate projection. (b) Distribution of output boxes from the first stage.

2) Network Architecture: Aligned with the oriented representation design, two-stage Ori-RCNN is chosen as baseline. The first stage retains the Oriented RPN to extract RRoI, while the second stage uses the two subheads of RoI-Trans for classification and precise OBB refinement. Additionally, to address CNN's limitation in capturing long-range dependencies within large and complex SAR images, the Swin transformer [36] is employed as the backbone of OSD for multiscale feature extraction. It distributes global information within scene slices using multihead attention.

The architecture of the improved network is depicted in Fig. 8. The backbone comprises four stages of feature extraction using a Swin transformer, each stage halving the feature map to generate multiscale features. These features are then merged using a feature pyramid network (FPN) and passed to the subsequent two-stage processing. Mirroring the approach of Ori-RCNN, the first stage uses oriented RPN to predict RRoIs, and the second stage crops these RRoIs to achieve oriented feature alignment. Then, the aligned features are processed through two convolutional layers, with the head network's classification and regression branches yielding category and bounding box location, respectively. The loss function of the network is defined as follows:

$$L_{\text{total}} = \lambda_1 \cdot L_{cls1} + \lambda_2 \cdot L_{\text{reg1}} + \lambda_3 \cdot L_{cls2} + \lambda_4 \cdot L_{\text{reg2}}. \tag{12}$$

Here, L_{total} denotes the combined multitask loss. L_{reg1} and L_{reg2} are previously delineated, targeting parameter offsets in the Mid-OR and the LSR. The classification losses for the first and second stages, L_{cls1} and L_{cls2}, employ a multiclass cross-entropy loss function, with the first stage being a binary foreground-background classification and the second stage involving multitarget classification, which is similar to binary classification in detection tasks. Additionally, λ_1, λ_2, λ_3, and λ_4 serve as balancing factors for each task's loss.

Additionally, to enhance model robustness in complex scenarios, strategic data augmentation strategies are leveraged, involving common data augmentation and multichannel feature fusion. Specifically, multifeature fusion is applied at the image level, including features from CFAR processing, 8 and 15 times the pixel mean contrast-stretching method.

D. False Alarm Discrimination Module (FDM)

After processing of CRM, the reduced detection scope lowers false alarm rates. Then, a false alarm discriminator is still important to address the residual false positives, primarily in sea-land isolation zones, maritime structures, and bridges. These arise due to insufficient learning of negative samples

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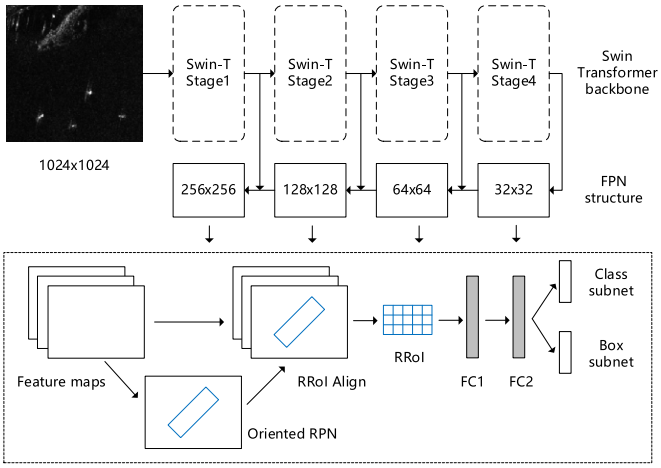


Fig. 8. Structure of OSD.

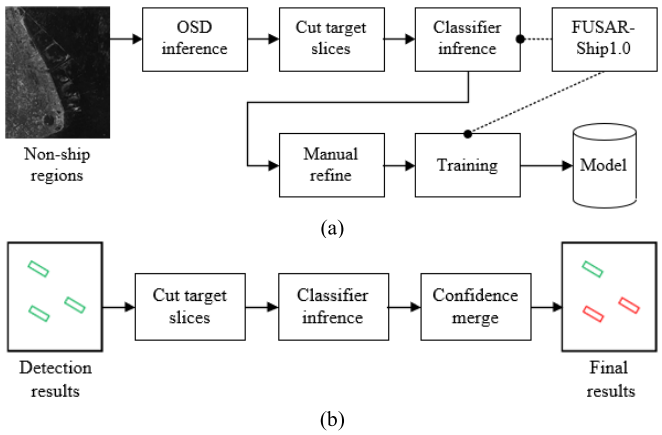


Fig. 9. Training and inference of discriminator. (a) Training process. (b) Inference process.

and the inclusion of nonship areas within candidate regions, as discussed in Section I.

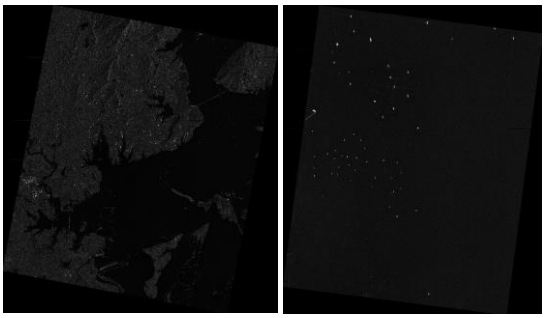
The FDM, structured as a deep learning classifier, is illustrated in Fig. 9, showcasing its training and inference process. To obtain effective false alarm samples for training, an initial detection round is conducted on all the original images from the training set. This process automatically segregates false alarm samples compared to ground truth, supplemented by false alarm data from the FUSAR-Ship1.0 [4]. Besides, the initially trained classification model on the FUSAR-Ship dataset facilitates efficient manual fine-tuning through preinference.

During inference, detection results are segmented and classified using the FDM. The final confidence score is a harmonized average of the classification and detection confidences, calculated as follows:

$$S_c = \frac{2S_d \times S_r}{S_d + S_r}. \tag{13}$$

Here, S_c is the combined confidence score, S_d and S_r are detection and discrimination confidence, respectively.

The FDM aims to accurately discern between ship and nonship targets, employing either binary or multiclass discrimination. This fine-grained approach to false alarm discrimination is essential due to the diverse appearance of false alarm



(a) (b)

Fig. 10. Examples of GF-3 whole-scene images in experiments. (a) Poyang Lake area, Jiangxi Province, China. (b) Bohai Sea area, China.

features; thus, FDM adopts a multiclass approach, categorizing seven target types: ship, land, coastal, wave, bridge, sea, and the other. The effects of category granularity and various deep learning networks on false alarm discrimination will be comprehensively analyzed in Section IV-D.

Additionally, similar to the detection network, the FDM also applies data augmentation and feature fusion to ship and false alarm target slices. The feature fusion integrates CFAR feature maps, 8× and 15× mean value stretched images.

IV. Experiments

A. Experimental Data and Metrics

The proposed method is employed on large-scene SAR imagery, and GF-3 images are used for experimental validation. GF-3 is China's first civilian fully polarimetric C-band SAR satellite, which is primarily used for marine remote sensing and monitoring. It boasts a variety of imaging modes and large imaging widths. The experimental dataset consists of 30 full-scene GF-3 images in the Ultrafine Strip (UFS) mode, featuring HH and VV polarizations and an approximate swath width of 30 km, characteristic of large scenes. Regarding resolution, the azimuth resolution is 1.124 m, and the slant range resolution varies from 1.700 to 1.754 m. After geocoding and geometric correction using commercial software, the images retain single-channel amplitude information, with dimensions around 30 000 × 34 000 pixels in width and height, covering an area of about one billion pixels. The imagery primarily covers regions with dense ship traffic, such as Hangzhou Bay, the Yangtze River Estuary, the East China Sea, and the Bohai Sea, featuring various complex maritime environments, including near-shore, off-shore, and alongside docking scenarios. Typical examples are shown in Fig. 10.

The dataset includes 3783 ships, divided into 20 scenes for training and ten for testing. For evaluation metrics, the CRM is evaluated using true hit rate (r_{hit}) and detection coverage (r_{cov}), as defined in Section III-B. Average precision (AP) and accuracy are used for OSD and FDM, with recall and false alarm rates evaluated on the whole-scene image results, defined as follows:

$$\text{AP} = \int_0^1 P(R)dR \tag{14}$$

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FN} + \text{FP} + \text{TN}} \tag{15}$$

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TABLE I
QUANTITATIVE EVALUATION OF SEA-LAND SEGMENTATION

	Sea-land isolation belt	r_{hit}	r_{cov}	Time /image
Train set		96.83%	75.52%	11.8s
	√	99.78%	77.36%	13.6s
Test set		96.21%	79.48%	12.2s
	√	99.61%	82.77%	13.3s

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{16}$$

$$\text{False alarm rate} = \frac{\text{FP}}{\text{TP} + \text{FP}}. \tag{17}$$

Here, TP, FP, TN, and FN denote true positive, false positive, true negative, and false negative, respectively.

Additionally, visualization of large-scene image detection results is presented using QGIS [37], and the configuration details of each module are provided in the subsequent experimental sections.

B. Evaluation of CRM

The CRM, as a nonlearning method, is tested for generalization ability by adjusting parameters manually based on the training set, followed by validation on the test set. The evaluation metrics include r_{hit}, r_{cov}, and process time. CRM involves two sequential processes: sea-land segmentation and CFAR processing. An initial evaluation of sea-land segmentation, as presented in Table I, reveals that the method employed reduces ineffective land detection areas by about 20%–25%. This reduction is primarily attributed to the presence of land regions in only some of the test images.

Additionally, the design of the sea-land isolation belt reduces missed detections of ships near coastlines. The morphological operations involved in the isolation belt take only about 1–2 s, indicating a minimal computational load. The mask matching approach, moreover, allows the algorithm to complete sea-land segmentation swiftly, enhancing its practical applicability.

Fig. 11(a) illustrates the visualization results of three scenes, each featuring extensive land areas. These results underscore the method's effectiveness in removing most land areas while preserving water bodies such as rivers and lakes. The application of sea-land isolation belt overcomes the geographic inaccuracies inherent in SAR imagery and mask-based matching, especially in coastal regions. Besides, Fig. 11(b) highlights a typical example of the sea-land isolation zone in an image. It reveals that many ships along the coast are effectively retained within the isolation belt. Additionally, as the isolation belt mainly appears in coastal areas and represents a minor proportion of the total area, its impact on the overall detection time is minimal.

Building on sea-land segmentation, the integration of CFAR preprocessing yields the conclusive result of CRM, detailed in Table II, with the processing time only accounting for CFAR. This approach narrows the detection scope, a result of the selected imagery's large oceanic areas and the effective elimination of most sea clutter by CFAR. Additionally, the coastline dilation enhances clutter modeling, thereby decreasing the missing detection of weaker ship targets. In terms of efficiency, the method's straightforward design, involving only global parameter estimation and threshold computation, ensures the module's promptness.

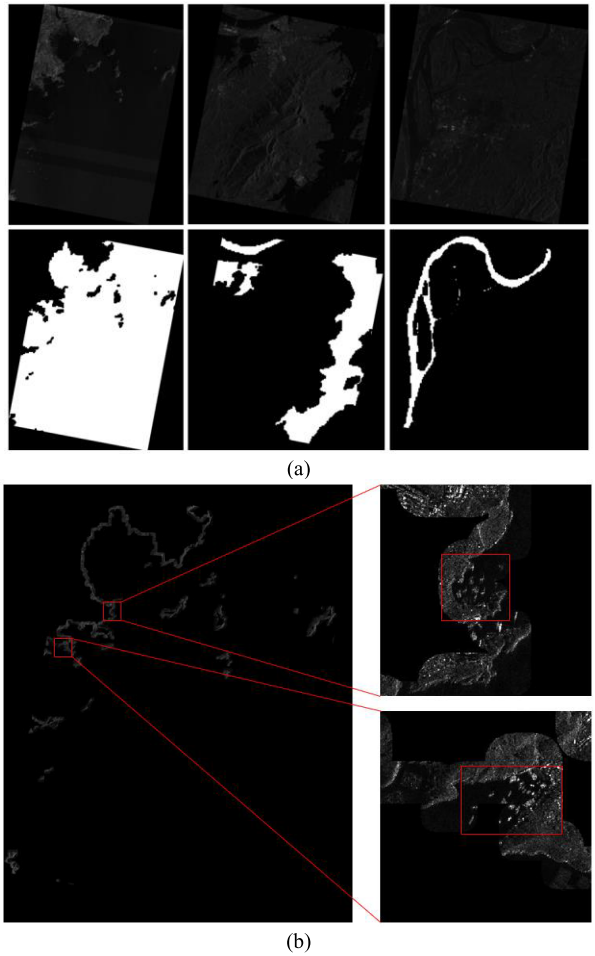

Fig. 11. Illustration of sea-land segmentation results. (a) Results of three images in the test set. (b) The sea-land isolation belt in one image of the test set.

TABLE II
QUANTITATIVE EVALUATION OF CFAR AND CRM

	Coastal line dilation	r_{hit}	r_{cov}	Time /image
Train set		98.91%	19.03%	6.5s
	√	99.64%	20.07%	7.3s
Test set		98.44%	19.22%	6.2s
	√	99.39%	20.29%	7.5s

Fig. 12 illustrates the visualization of one scene's result, showcasing the output of CRM. Enlarged views of two marine areas highlight the efficacy of CFAR in removing sea clutter while retaining the pixels of ships across various scales.

C. Evaluation of OSD

In this section, the proposed OSD is compared with the current state-of-the-art (SOTA) algorithms, including various single-stage and two-stage methods. The experimental data

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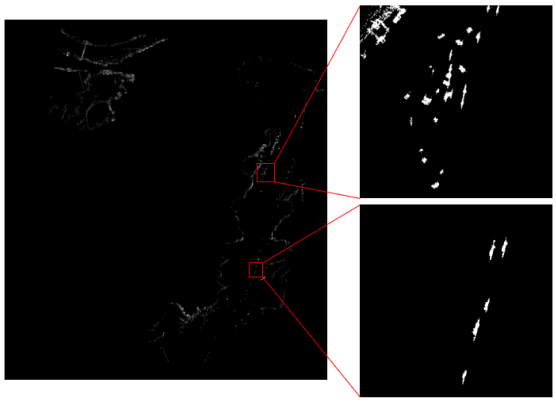

Fig. 12. Illustration of CFAR effect and final result for CRM.

consists of scene slices containing ship areas from whole images, each slice being 1024 × 1024 pixels. A total of 2555 training slices with 7447 targets from 20 training scenes are involved, while the test set comprises 1037 slices with 2680 targets from ten test scenes. All algorithms employ the FPN with an input size of 1024 × 1024 pixels. The model training is conducted on four RTX 3090 GPUs over 36 epochs with a BS of 8. The inference is carried out on a single GPU with a BS of 1. The confidence threshold for the results is set at 0.05. Evaluation metrics include AP, frames per second (FPS), and GPU memory (Mem.) usage during inference. Controlled variables in the experiments included the implementation of data augmentation.

As illustrated in Table III, the results reveal that single-stage methods generally offer faster processing but fall short in overall performance compared to two-stage methods. Among the single-stage methods, KLD [23] stands out for its superior performance, using Kullback–Leibler divergence for box regression to effectively alleviate the PoA issue. Incorporating the R3Det [19] structure enhances KLD's detection efficacy, but the cascaded network introduces extra inference time. Ori-RCNN, employing a two-stage structure, adds one convolutional layer in the RPN stage to generate RRoI, attaining notable levels of detection accuracy and time efficiency. The proposed OSD, advancing over Ori-RCNN, further improves detection capabilities, achieving the highest AP. Despite a small decrease in speed due to network enhancements, the algorithm still performs at a high level, balancing speed with precision.

Subsequently, the strategic data augmentation, inclusive of generic approaches and multichannel feature fusion, is integrated into the most optimal single-stage and two-stage algorithms, KLD and Ori-RCNN, as well as the proposed OSD. The results reveal that the strategy enhances performance across all detectors. For OSD, it led to a 1.8% increase in AP, outperforming the other methods.

Fig. 13 displays a subset of the visualized results for KLD, OSD, and the implemented data augmentation strategy. This visualization highlights OSD's enhanced capability in accurately predicting the orientations of ships in areas of dense ship arrangements, as exemplified in the first column. The data augmentation enhances detection performance across various complex scenarios. Besides, the ships depicted in the third and fourth columns have weaker energy and less distinct features, and data augmentation enhances the detectability of these targets.

TABLE III
QUANTITATIVE EVALUATION OF OSD AND SOTAS

	Type	Data Aug.	AP (%)	FPS	Mem. (GB)
RetinaNet-R	One-stage		0.861	21.6	2.14
R3Det			0.886	16.7	2.17
GWD			0.879	20.2	2.14
RetinaNet-KLD			0.883	21.2	2.14
R3Det-KLD			0.892	16.7	2.17
		√	0.884		
RoI-Trans	Two-stage		0.86	19.1	2.8
Gliding Vertex			0.849	21.1	2.64
Ori-RCNN			0.899	20.5	2.69
		√	0.903		
Proposed OSD			0.911	17	2.46
		√	0.929		

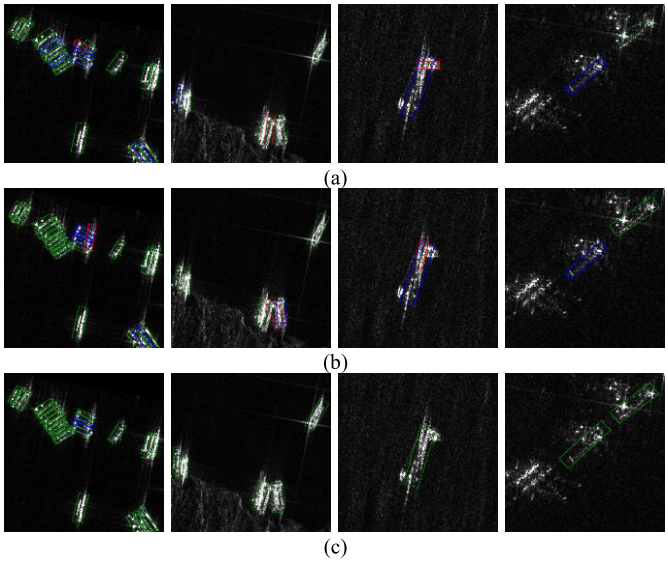

Fig. 13. Visualizes results from KLD, OSD, and the data augmentation strategy; green, blue, and red indicate correct detections, missed detections, and false alarms, respectively. (a) KLD. (b) OSD. (c) OSD with data augmentation.

D. Evaluation of FDM

The proposed FDM employs a multiclass approach for discerning false alarms, including ship and six categories of false alarms. This Part initially explores the impact of granularity in false alarm categorization on the discriminator's effectiveness. The experiments use VGG16 as a classifier, with 20% of the FUSAR-Ship false alarm database set aside as a validation set. Three levels of granularity are compared: binary classification (ships versus all false alarms), five-category classification (ships, land and coastal, sea, bridges, and others), and seven-category classification (all categories).

As shown in Table IV, where nonship accuracy considers all false alarms as a single negative class, multiclassification outperforms binary classification, with finer categorizations

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TABLE IV
EVALUATION OF FINE-GRAINED FALSE ALARM CLASSIFICATION

Category number	Overall accuracy (%)	Ship accuracy (%)	Non-ship accuracy (%)
2	96.31	95.87	97.69
5	98.12	99.03	97.22
7	98.46	99.62	97.43

TABLE V
EVALUATION OF RESULTS ACROSS DIFFERENT CLASSIFICATION NETWORKS

	Data Aug.	Overall accuracy (%)	Flops (G)
VGG16		96.71	15.5
	√	98.46	
ResNet50		97.12	25.56
	√	99.32	
Swin-tiny		97.84	28.29
	√	99.45	

leading to higher accuracy. With binary granularity, the diversity of nonship false alarms is highest, resulting in better nonship accuracy but lower ship detection rates. Higher ship accuracy, indicative of fewer missed ship detections, is a key focus for auxiliary false alarm discriminators. This study, therefore, adopts the optimal seven-category strategy.

This section, furthermore, compares the performance of different DL networks and the effectiveness of data augmentation. The experiments, using the seven-category granularity, evaluate overall accuracy and time efficiency in terms of floating-point operations per second (Flops), where G represents 10^9 operations. The comparison includes VGG16, ResNet50, and Swin-transformer tiny (Swin-tiny). According to Table V, data augmentation enhances the classification performance of all three networks, validating the effectiveness of the augmentation strategy. While ResNet and Swin-tiny are more complex and slower than VGG16, they achieve higher accuracy. Consequently, ResNet50 is selected as the discriminator in FDM.

E. Ablation Study Based on Full-Scene Images

Ablation experiments are conducted to evaluate the individual contributions of each component in large-scene SAR images. The OSD is trained using four RTX 3090 GPUs, with BS of 8, an initial learning rate of 0.0001, over 72 epochs, and included data augmentation. Similarly, the FDM training used four GPUs, with BS of 64 and an initial learning rate of 0.01, for 100 epochs, also integrating data augmentation. For whole-scene inference, a single GPU is employed, with each module processed sequentially and the BS for OSD and FDM set at 1 and 64, respectively. Evaluation metrics focused on recall, false alarm rate, and inference time for comprehensive whole-scene image analysis.

While the efficacy of the OSD has been previously validated in Section IV-C, this ablation study centers on the CRM and FDM, using the OSD as a baseline. The quantitative evaluations are detailed in Table VI. Visualization results, with a confidence threshold of 0.8, are depicted in Fig. 14. The figure illustrates detection results for three full-scene images and the impacts of the individual modules in specific areas.

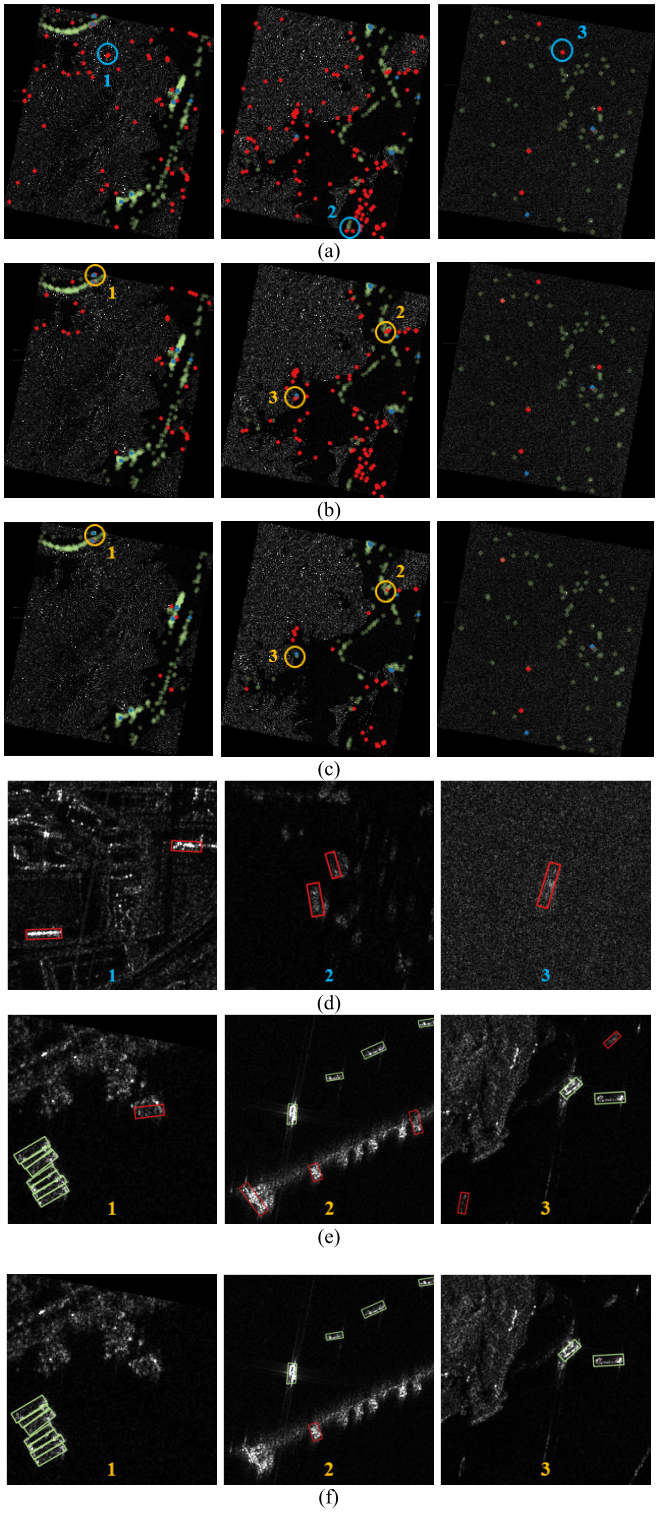

Fig. 14. Whole-scene visualization of ablation study results; green, blue, and red indicate correct detections, missed detections, and false alarms, respectively. (a) The baseline. (b) With the addition of the CRM. (c) Incorporating both CRM and FDM. (d)–(f) correspond to magnified displays of the three circled areas in (a)–(c), respectively.

From the first row of Table VI, it is evident that employing only OSD resulted in a long inference time and a high false

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TABLE VI
ABLATION STUDIES ON CRM AND FDM

CRM	FDM	Recall (%)	False alarm rate (%)	Average time (s) / image
		92.21	30.76	431
√		92.18	23.57	82
√	√	91.47	12.69	96

alarm rate, making it unsuitable for practical use. Fig. 14(a) illustrates plenty of false alarms in inland and sea surface areas. The three regions highlighted by blue circles are magnified in Fig. 14(d), showing false alarms caused by inland buildings, small islands, and sea clutter.

Incorporating the CRM improves efficiency, reducing both inference time and the false alarm rate. Comparative analysis of full-scene results in Fig. 14(a) and (b) shows effective filtering of most land-based and oceanic false alarms; however, some false alarms in coastal and sea clutter areas persist, as OSD alone struggled to filter out all false alarms look like ship targets, as shown in Fig. 14(e).

The integration of the FDM, as indicated in the third row of Table VI, slightly increased the overall detection time but further decreased the false alarm rate. As seen in Fig. 14(b) and (c), this results in fewer coastal nonship false alarms but also a minor decrease in the detection rate, primarily due to misclassifications by the discriminator. Fig. 14(e) and (f) illustrates the areas marked by three yellow circles in (b) and (c), highlights the module's effectiveness in further filtering false alarms from ocean, artificial structures, and coastal areas that are retained post candidate region retrieval.

With all modules combined, the algorithm, furthermore, averaged 96 s for each full-scene image. This efficiency demonstrates minute-level detection capability on images encompassing around one billion pixels.

The integration of the FDM increases the complexity of the proposed method. Currently, mainstream detectors are end-to-end, capable of simultaneously performing detection and classification, such as the OSD. In this context, the proposed method is also compared with an end-to-end detection approach. For the end-to-end method, false alarm annotations are retained at the scene scale and trained together with ship categories in OSD. Then, the predicted false alarms are directly discarded during inference. The experimental results are shown in Table VII. "One-class labeled for OSD" indicates that all false alarms are marked as one category, while "Five-class" means the false alarm categories are marked as the same five categories used in FDM. Results demonstrate that FDM surpasses the end-to-end method in recall and false alarm rate, albeit with a slight increase in detection time; however, it still maintains a rapid level for large-scene images. The potential reasons for this are twofold.

1) FDM also uses false alarm sample slices from the FUSAR-Ship1.0 dataset, which are not directly applicable to the end-to-end detection network.

2) Marking false alarms results in many training slices without ship targets, exacerbating the imbalance of positive and negative samples during the detector's learning process.

TABLE VII
COMPARISON OF FALSE ALARM SUPPRESSION STRATEGIES

	Recall (%)	False alarm rate (%)	Avg. Time (s) / image
One-class labeled for OSD	90.15	16.85	84
Five-class labeled for OSD	90.46	15.62	83
FDM	91.47	12.69	96

TABLE VIII
COMPARATIVE EXPERIMENT RESULTS

	CRM &FDM	Recall (%)	False alarm rate (%)	Avg. Time (s) / image
Faster-RCNN		84.87	45.50	227
	√	84.26	25.56	85
CenterNet2		85.79	56.89	217
	√	85.18	33.68	76
R3Det-KLD		89.34	55.93	311
	√	88.83	20.02	104
Oriented R-CNN		89.04	38.02	312
	√	88.63	11.28	93
The proposed method	√	91.47	12.69	96

The end-to-end detection of targets in large-scene SAR images, however, remains a topic worthy of further study. Future research will mainly focus on how to balance positive and negative samples and on data augmentation based on target-level slice samples.

F. Comparative Evaluation With Other Methods

In this section, the proposed method is experimentally compared with the SOTA detectors. The comparative methods include HBB detectors (Faster-RCNN, CenterNet2) and OBB detectors (R3Det-KLD, Ori-RCNN). Faster R-CNN, a classic two-stage detector, is widely used in remote sensing target detection, while CenterNet2 represents a typical example of anchor-free detectors. Additionally, R3Det-KLD and Ori-RCNN are among the best in single-stage and two-stage OBB detectors, respectively. The detection performance is evaluated on whole-scene images; furthermore, ablation experiments testing the proposed CRM and FDM are also conducted on comparative methods.

According to Table VIII, the proposed method significantly surpasses other methods in detection speed and false alarm rate reduction. While other methods average around 4–5 min per scene for detection, the proposed method requires only 96 s. The false alarm rates of other methods exceed 30%, whereas the proposed method maintains it at 12%. In terms of detection rate, faster-RCNN and CenterNet2 are relatively weaker, whereas KLD and Ori-RCNN are closer to the proposed method.

The advantages in speed and false alarm suppression are primarily attributed to the CRM and FDM. Incorporating these modules into the comparative algorithms significantly boosts their performance, nearly matching the detection speed of the proposed method. Their detection rates and false alarm rates,

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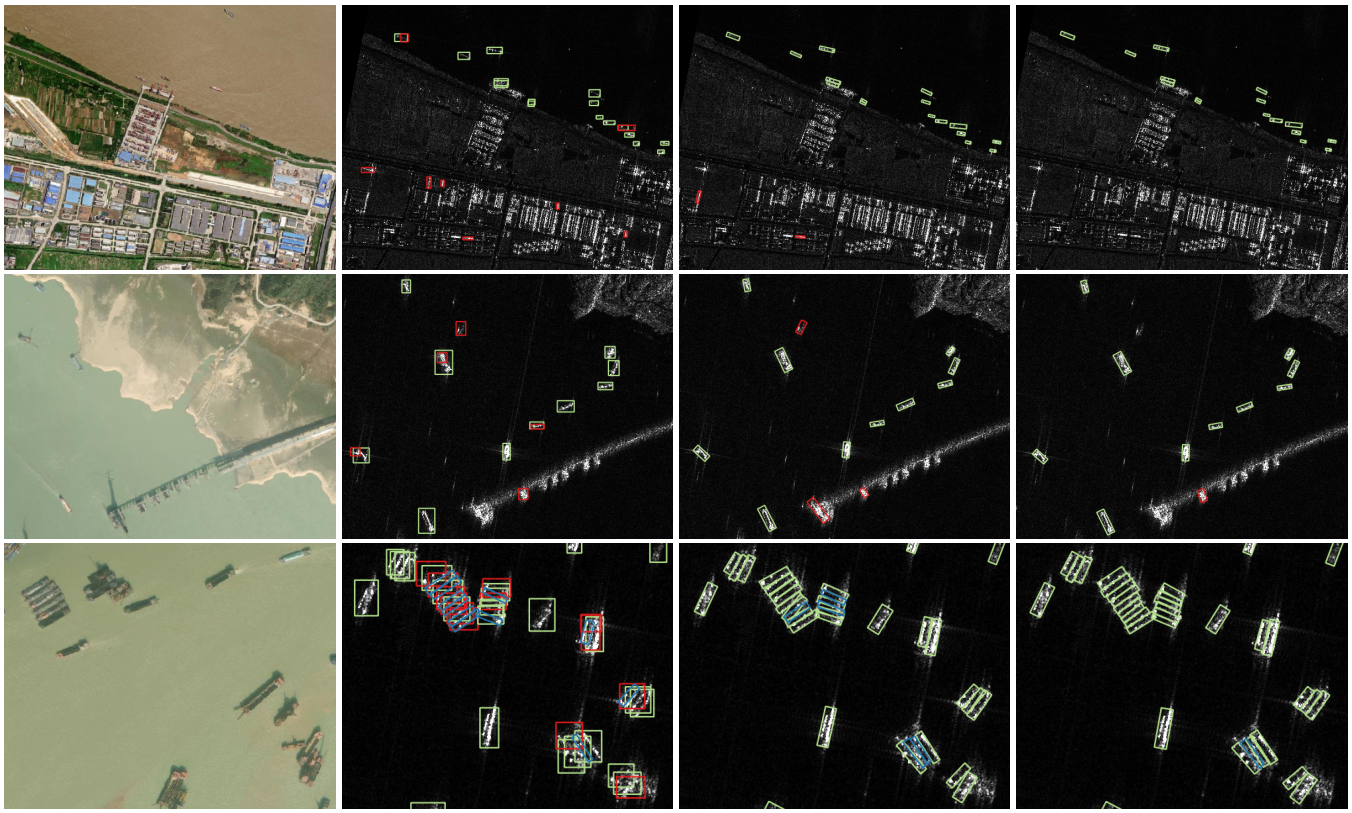

Fig. 15. Visual comparison between the proposed method and other methods; green, blue, and red indicate correct detections, missed detections, and false alarms, respectively. (a) Corresponding optical image. (b) CenterNet2. (c) Oriented R-CNN. (d) The proposed method.

however, remain inferior. Notably, Ori-RCNN has a slightly lower false alarm rate than the proposed method but lags by 3% in detection rate. Overall, the proposed method achieves optimal detection results while maintaining a comparatively swift processing speed.

Fig. 15 compares the visual results of CenterNet2, Ori-RCNN, and the proposed method, covering near-shore, off-shore, and densely packed ship scenarios. It is evident that, compared to other methods, the proposed approach more effectively suppresses false alarms in various scenarios, such as land-based false alarms in the first row and sea surface false alarms in the second row, primarily arising from man-made objects and SAR-specific noises. Besides, CenterNet2, being an HBB detector, displays a comparable recall for off-shore ships but exhibits a higher miss rate in scenarios involving densely packed near-shore ships, culminating in a reduced overall detection rate. In contrast, Ori-RCNN and the proposed method, employing oriented detection structures, exhibit fewer misses.

Fig. 16 presents the detection results of the proposed method on a test image, highlighting four different maritime scenarios: near-shore, off-shore, and densely arranged areas. This visualization shows our method's proficiency in accurately detecting ships across diverse and complex sea areas. Nonetheless, instances of missed detections and false alarms still remain. For example, the top-right segment of the image illustrates false alarms near a coastal dock and missed detections among densely clustered near-shore ships; moreover, a missed detection of a small-sized, densely packed ship is evident in the bottom-left region of the image. These ships are either less distinct or considerably small, pinpointing areas for future research enhancement.

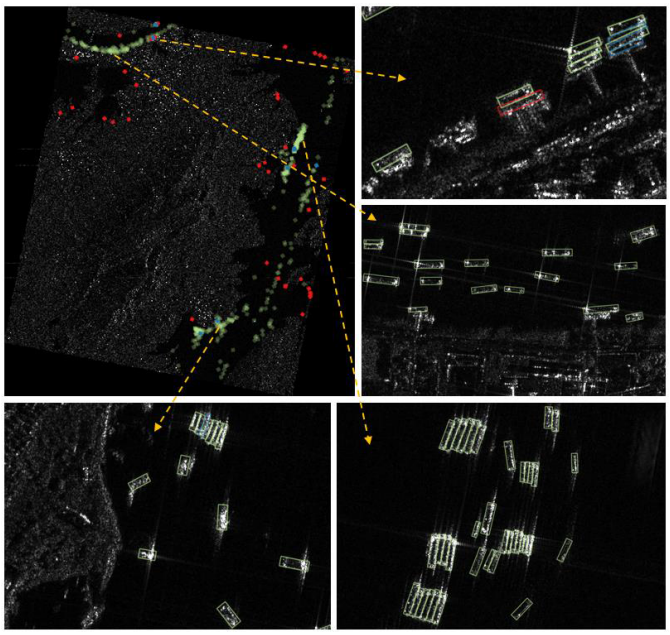

Fig. 16. Detection results of the proposed method in a selected whole-scene image; green, blue, and red indicate correct detections, missed detections, and false alarms, respectively.

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V. Conclusion

The proposed method addresses the efficiency of ship target detection in very large, full-scene, single-channel SAR imagery. It introduces a multiscale, progressive detection method. This approach integrates global-scale candidate region retrieval, scene-scale oriented ship detection, and target-scale false alarm filtering, aiming for rapid and precise ship detection. It presents a comprehensive framework, consisting of CRM, OSD, and FDM. The CRM is designed using traditional approaches, incorporating sea-land segmentation and CFAR processing for fast localization of potential ship areas. Subsequently, DL-driven OSD and FDM discern ships from these candidate regions. The efficacy of each module is validated through individual experiments, and the method's overall effectiveness in processing full-scene SAR imagery is demonstrated via ablation and comparative studies.

VI. Declaration of Interest Statement

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

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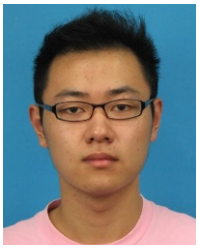

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